

## Stabilised Platform vs Strapdown.

### Stabilised platform:

- \* the inertial sensors are placed on an inertially-stabilised platform mounted on the inner gimbal of a 3-axis gimble set.
- \* the gyros measure any angular rotation of the platform which is used as a feedback signal to control torque motors on each gimble axis.
- \* Platform attitude is measured by resolvers on each gimble axis.

### Strapdown equation:

- \* gyros and accelerometers are rigidly mounted to the vehicle
- \* therefore, measure the specific force and angular velocity of the vehicle on the body axes.

### Stabilised platform CONS:

- mechanically complex
- costly to manufacture/maintain
- larger volume/weight.
- cannot be used directly for airframe stabilisation.

Strapdown: much more computationally complex.